

STUDY GUIDE UNDP



Agenda 1 : Strengthening Mechanisms
for the Inclusion of Marginalised Groups
in Development Programmes

Agenda 2 : Addressing Global
Challenges in Water Governance





Letter from R&D

Dear Delegates,

The Department of Research & Development of MUNSociety MPSTME welcomes you to the 11th edition of Mumbai MUN! Our department upholds a proud tradition of facilitating meaningful discourse on national and international matters, encompassing both historical events and current issues.

We are primarily tasked with formulating agendas and preparing comprehensive study guides. We have created detailed and engaging guides aimed at providing valuable resources for all delegates.

This year's theme, "Redefine the Norm," challenges us to think beyond conventional solutions and embrace innovative approaches to global challenges. In a rapidly evolving world, traditional methods often fall short of addressing contemporary issues. We believe that transformative change requires the courage to question established practices and the creativity to envision alternative pathways.

With this theme in mind, we have thoroughly picked agendas and committees that demand fresh perspectives, challenge conventional thought, and encourage innovative approaches to problem-solving. In this study guide, you will find well-researched background information to help you prepare for your committee sessions.

We, as facilitators, aim to provide you with the intellectual arsenal to combat any challenges that you might face and make the most of your Mumbai MUN experience.

We look forward to your active participation and contributions during Mumbai MUN and are confident that, together, we can make this conference a memorable and enriching experience for all.

Best wishes,

Department of Research & Development,
MUNSociety MPSTME.

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COMMITTEE OVERVIEW

The United Nations Development Programme (UNDP), established in 1965 by the UN General Assembly, was created by integrating the United Nations Expanded Programme of Technical Assistance (1949) and the United Nations Special Fund (1958). It promotes change that helps people build a better life, and is primarily concerned with eradicating poverty, building democratic governance, rule of law and inclusive institutions.

By the 1970s the UNDP expanded its role to not only include economic growth but also social equity, environmental sustainability, and human development. The first Human Development Report was published in 1990, introducing the Human development Index(HDI).

By the 2000s, UNDP played a pivotal role in the formulation of the Millennium Development Goals(MDGs), a set of eight global goals aimed at reducing extreme poverty and improving health and education by 2015. These acted as the precursor to the Sustainable Development Goals(SDGs) which will be delved upon later in this document.¹

UNDP's headquarters are located in New York. UNDP has representation offices in Geneva, Brussels, Copenhagen, Tokyo, and Washington D.C. Further, UNDP has Regional Hubs in Addis Ababa, Amman, Bangkok, Istanbul and Panama City.

MANDATE

The UNDP's mandate is fundamentally anchored in the pursuit of the Sustainable Development Goals (SDGs), addressing critical global challenges such as poverty reduction, climate action, gender equality, and resilience building.

Central to its mission is the commitment to improve the lives of marginalised communities, ensuring that no one is left behind in the development process.²

The UNDP emphasises the importance of sustainable economic development across various fronts. By focusing on local capacity building, the agency strives to empower communities, fostering long-term self-sufficiency and prosperity. This involves developing tailored solutions that reflect the unique cultural and socio-economic contexts of different regions. Through initiatives that enhance education, vocational training, and entrepreneurship, the UNDP enables individuals to contribute effectively to their local economies.

Moreover, the UNDP's global footprint allows it to understand and respond to crises effectively. Whether addressing the impacts of climate change or responding to humanitarian emergencies, the agency remains agile and adaptive, tailoring its interventions to meet urgent needs while also prioritising sustainable outcomes. In this way, the UNDP not only supports immediate recovery efforts but also lays the groundwork for resilient communities capable of withstanding future challenges.

With a presence in over 170 countries, the UNDP collaborates closely with governments, civil society organisations, and the private sector. This collaborative approach facilitates the development of customised strategies that address specific regional challenges. By leveraging the knowledge and resources of diverse stakeholders, the UNDP fosters an environment where innovative solutions

¹United Nations Development Programme: Home. <https://www.undp.org/>

²UNDP. Strategic Plan 2022–2025. <https://strategicplan.undp.org/>

Agenda 1: Strengthening Mechanisms for the Inclusion of Minority Groups in Development Programmes

INTRODUCTION

Understanding Minority Groups

There are many subjective and objective elements involved with the term 'Minority', and there is no internationally agreed definition on the term. In their definitions, the UN has focused on national, ethnic, religious and linguistic groups that typically constitute less than half of the population in the territory of a State. It is, however, understood that these categories may overlap in many cases, and are not definite. For example, although refugees and stateless persons may not fall under the mentioned categories, they may still be considered as minority groups in the States in which they settle.

These may also include persons of disability, alternately referred to as the "world's largest minority", and persons of different gender identity and sexual orientations. While there are mandates and mechanisms in place for protecting the rights of the indigenous people, they can also claim minority rights under international law. Both parties wish to retain and promote their identity.

According to the 2011 census, over 201 million persons were identified as belonging to Scheduled Castes, and 101 million to Scheduled Tribes in India whereas China in 2010 identified 113 million as ethnic minorities. More than a third of the US population identify as Black, Asian, Native American, Hawaiian, Hispanic, and others. In 2016. Thus, different definitions and data collection methods provide different minority estimates.

Regardless of the varied understanding of the term 'minority', most minority groups typically

face different forms of discrimination that lead to their marginalisation and exclusion. Many are deemed "undeserving" by different parties as they are not seen as "traditional" minorities, which leads to their exclusion. For instance, there may exist instances wherein groups within the minority, such as women and children, the elderly and persons living with HIV are marginalised. This multi-dimensional discrimination observed may lead to exclusion and dissociation within the community and need to be accounted for.

In order for the minority groups to contribute towards the social and political stability of the society in which they live, it is essential to adopt their rights as human rights; and provide adequate protection and promotion. Among the different UN international human rights instruments, the 1992 United Nations Minorities Declaration is the only one that is solely devoted to minority rights. It was inspired by Article 27 of the International Covenant on Civil and Political Rights.³

Importance of Minority Inclusion in Development Programmes

One of the major factors in poverty and inequality is discrimination against the minority. There can be significant benefits to overcoming the marginalisation of minorities in national economies. Better attention to the case of minorities by the government can promote inclusion and human development in a sustainable manner.

The need for social protection was addressed in The World Summit for Social Development, and it was only in the 2000s that the concept of social

³"Minorities." *United Nations*, www.un.org/en/fight-racism/vulnerable-groups/minorities.

protection as a policy instrument started gaining momentum.

The eight Millennium Development Goals (MDGs) were commitments made by the governments and the international community as a global effort to address poverty, education to all children and deadly disease prevention. It led to over a billion people being pulled out of extreme poverty, lowered child mortality and school dropout rates since 1990. Its universally agreed objectives and achievements paved the way for the Sustainable Development Goals (SDGs) which was adopted in 2015. It is recognised that 17 SDGs help with the universal call to action to further eradicate poverty, protect the planet and promote peace and prosperity by 2030.

In a bid to alleviate the far-reaching impact of the 2008 financial crisis, the Social Protection Floor initiative was established in 2009, which was endorsed by Member States at the 2012 United Nations Conference on Sustainable Development (Rio+20). Social Protection as a key policy tool was further reinforced by the adoption of the 2030 Agenda, and the SDG goals directly concerning the issues of the minority are SDG 1: No Poverty, SDG 3: Good Health and Well-Being, SDG 5: Gender Equality, SDG 8: Decent Work and Economic Growth and SDG 10: Reduced Inequalities. It has been universally accepted that actions in one area can positively reinforce the others and contribute to social, economic and environment sustainability.^{4 5}

CHALLENGES FACED BY MINORITY GROUPS

Socio-Economic Status and Education Disparities

The socio-economic status of the minority is not only concerned with income, it also includes education, overall health, financial security and all factors that contribute to a better quality of life. According to research conducted in 2016, multidimensional poverty is more likely to be faced by minority racial groups than their White counterparts. Although the high school dropout rates have narrowed over the years since 2000, the Latinos remain the highest followed by the African-Americans according to the National Center for Education Statistics, 2015.

Health Disparities

Racial and ethnic minorities face more problems due to factors such as the inability to navigate the system, inadequate access to quality healthcare, provider ignorance or bias and stress. The different levels of discrimination has also shown to contribute to the mental health disorders among Asian Americans and African Americans.

Terrorism

Rooted in ethnic and religious grievances, a great source of insecurity in the world is terrorism. States can utilise the instability to persecute minorities in the name of 'terrorism'. A direct threat to minorities can also be global militarism, as with the Ilois people who are unable to return to Chagos archipelago due to the US base there.

⁴Promoting Inclusion through Social ..., www.un.org/development/desa/dspd/wp-content/uploads/sites/22/2018/06/rwss2018-full-advanced-copy.pdf.

⁵"Sustainable Development Goals." UNDP, www.undp.org/sustainable-development-goals.

Threats to Minority Rights and International Standards

In certain countries, minority rights itself is under threat alongside human rights. Often, these aren't acknowledged by the NGOs and denied by the governments. The aim to develop a legally binding international standard has stalled since the 1992 Declaration on the Rights of Persons Belonging to National or Ethnic, Religious or Linguistic Minorities.

With limited developments being made in Europe over the last few years, the European Convention on Human Rights has no minority rights protocol. Although some European countries such as Hungary, Austria and Finland have taken up the issue of international minority rights, the countries with direct interests tied with minority protection have hardly acknowledged the issue. There is a lack of understanding among policy-makers regarding their rights. It is necessary for the national and international laws to successfully implement the rights of minorities, and the minorities to be able to use them to address their concerns.⁶

Inadequate Data Collection on Minority Needs

Effective inclusion in development programmes is challenged by a lack of disaggregated data on minority groups. The United Nations Department of Economic and Social Affairs (UNDESA) has noted that insufficient data limits the ability to design targeted interventions for minorities. Without comprehensive data that includes variables like ethnicity, language, religion, and geographic location, development programs cannot effectively address the specific barriers minorities face.

Resource Allocation and Funding Constraints

Ensuring that minority-focused initiatives receive adequate funding remains a significant challenge. Reports from the UNDP suggest that development projects often lack designated funding for minority inclusion. When resources are insufficient or unevenly distributed, programs may only partially benefit minority groups, which limits their overall effectiveness in addressing inequalities.

Barriers to Participation in Decision-Making Processes

Minorities often lack representation in development programs, making it difficult for them to influence policies that impact their lives. International agreements, like Article 25 of the ICCPR and Article 2.2 of the UN Declaration on Minority Rights, affirm their right to participate in public and cultural life.⁷ However, discrimination, citizenship restrictions, and political exclusion frequently prevent meaningful participation, resulting in development strategies that may overlook their specific needs.

MINORITY RIGHTS UNDER INTERNATIONAL HUMAN RIGHTS LAW

When it comes to the rights of minorities, the main point of reference for the international community is the United Nations Declaration on the Rights of Persons Belonging to National or Ethnic, Religious and Linguistic Minorities. According to the declaration, persons belonging to minorities have the right to enjoy their own culture, to profess and practise their own religion

⁶"Ethnic and Racial Minorities & Socioeconomic Status." *American Psychological Association*, www.apa.org/pi/ses/resources/publications/minorities.

⁷*Marginalised Minorities*. (n.d.). ohchr. <https://www.ohchr.org/sites/default/files/Documents/Issues/Minorities/UNDPMarginalisedMinorities.pdf>

and the right to use their own language. In order for the persons to be able to enjoy such rights, the declaration contains measures that the States should implement. This includes promoting and educating the broader public of the history, culture and traditions of the minorities existing within their territories. The States should also create an environment where the persons belonging to minorities are able to fully participate in the development and economic progress of their country.

The contents of the declaration was inspired by article 27 of the International Covenant on Civil and Political Rights. The International Covenant on Economic, Social and Cultural Rights mentioned explicitly that all parties agree that the rights listed in the declaration are applicable for everyone regardless of their race, sex, language, religion or any such discrimination.

Other major sources of minority rights include article 1 of the International Convention on the Elimination of All Forms of Racial Discrimination and Article 30 of the Convention on the Rights of the Child.

The rights of the minority are human rights with four major concerns: The protection of their existence; The protection and promotion of their cultural and social identity; Fostering non-discrimination and equality thus putting an end to systemic discrimination; and effective participation of the members with regards to decisions that affect them.^{8,9}

UN STANDARDS AND MECHANISMS

Different committees have been established to ensure the rights written in the international human rights conventions. Some of these that are directly relevant to the rights of the minority include the Human Rights Committee; The

Committee on Economic, Social and Cultural Rights; The Committee on the Elimination of Racial Discrimination; The Committee on the Rights of the Child; The Committee on the Elimination of Discrimination against Women and The Committee on the Rights of Persons with Disabilities.

Periodic reports are submitted by the States to these committees which in turn submit observations to see if the measures taken up by the States ensure the rights of the minorities. In case of any violations of any rights, the committee urges the State to refrain from any further violations and/ or provide measures to rectify the situation.

Early Warning Mechanisms

Early warning mechanisms also exist, like with the Committee on the Elimination of Racial Discrimination which brings to attention situations where alarming levels of discrimination has been reached, and urgent and effective response is needed from the concerned States. A similar early warning mechanism exists for the Secretary-General and Security Council so they may be able to promptly respond to situations that could lead to genocide, as per the mandate of the Special Adviser on the Prevention of Genocide, 2005.

Other Mechanisms

Other mechanisms include the United Nations Independent Expert on minority issues (2005) which works on the promotion of the United Nations Minorities Declaration. Its mandate also includes identifying the best practices with the technical cooperation by OHCHR, consultation with the Governments regarding their minority issues and taking into consideration the views of the non-governmental NGOs.

⁸Minority Rights: www.ohchr.org/sites/default/files/Documents/Publications/MinorityRights_en.pdf

⁹"Minorities." United Nations, www.un.org/en/fight-racism/vulnerable-groups/minorities.



United Nations Forum on Minority Issues
Source: Federal Union of European Nationalities

United Nations Human Rights Council

The Human Rights Council established the Forum on Minority Issues in 2007, as a platform for the cooperation on issues relating to national, ethnic, religious and linguistic minorities. It provides thematic funding, a solution made to maximise donation impacts made by the United Nations Children's Fund (UNICEF)¹⁰. The forum also works with independent experts on minority issues to utilise their analysis to identify and adopt the best practices and initiatives for future implementation of the United Nations Minorities Declaration. This expert also writes reports on the annual meetings that the forum prepares. It is also expected of the forms to contribute to the High Commissioner for Human Rights efforts to improve cooperation among the UN bodies, agencies and mechanisms and their programs and activities on the protection and promotion of minority rights.

The Office of the High Commissioner for Human Rights

The Office of the High Commissioner for Human Rights (OHCHR) gives secretariat support to different special procedures to work on the examination, monitoring, advising and the

public reporting on human rights situations. They work on responding to individual complaints, conducting studies, providing advice and engaging in promotional activities. "Country mandates" are those responsible to specific countries and territories and "thematic mandates" are those working on specific worldwide human rights concerns.

These mandate-holders collect information on human rights violations, sending appeals to governments for clarification and doing country visits when invited by the government.

Besides the independent experts on minority issues, other experts focus on minority rights, such as extrajudicial executions, and basic needs from food, shelter, health and education to rights relating to expression and religion, and also protection against racism, torture, trafficking, and violence. New mandates, like the independent expert on cultural rights, are also relevant to minority concerns.¹¹

United Nations in Zambia

The Government of Zambia and the United Nations in Zambia collaboratively initiated a Programme of Sustainable Resettlement, designed to align with the 2030 Agenda for Sustainable Development.¹² This program supports the long-term integration of former Angolan and Rwandan refugees into designated Zambian communities, facilitating their gradual settlement and adaptation within local society. Additionally, it emphasizes the inclusion of youth in the Mwange resettlement area, recognizing that meaningful integration requires careful, sustained planning over time. The United Nations Development Programme (UNDP) leads this initiative, underscoring the importance of a

¹⁰"Thematic Funding." UNICEF, www.unicef.org/partnerships/funding/thematic-funding

¹¹Minority Rights, www.ohchr.org/sites/default/files/Documents/Publications/MinorityRights_en.pdf

¹²Inclusive growth. (n.d.). United Nations Development Programme. <https://www.undp.org/zambia/inclusive-growth>

strategic, gradual approach to successful resettlement.



Programme of Sustainable Resettlement

Source: UNDP

CASE STUDIES

Vietnam

Vietnam has 54 ethnic groups with their own specific cultures, lifestyles and languages. Over 86% of the total population is Kinh people with the rest like Tay, Thai and other ethnic groups collectively covering 10% of the total population. Most of these ethnic groups especially the ones with small populations live in the highlands and mountainous areas almost cut off from infrastructure, health care and education. Even after the recent rapid economic growth poverty is still prevalent with as high as 50% of the total poor population coming from the 14% Ethnic Minorities (EMs).

To tackle this, the government introduced the National Target Programme on Sustainable Poverty Reduction in 2012 and many other international organisations also followed suit. This however was not able to eradicate the chronic poverty which was still present especially in the northern mountainous areas. Many studies have been conducted over the years and the general consensus is that the issue is the barriers in accessing resources needed for productive activities, like education, capital, markets, and agricultural land. This stems from the long geographical distance and the poor quality of the

services like education and health near their homes. The hygiene, water conditions and housing are also generally poorer than those of relatively larger ethnic groups like Kinh.

It is also seen that while the average life expectancy of EMs is 72.1 years. The EMs with poor access to good healthcare and hygiene conditions like La Hu, Lu, Chut and others, their life expectancy ranged around 62-65 years. It is also seen that the EMs have high proportions of married people (52% and above) and while the average national average age of marriage is around 25 years, and that of the EMs is significantly lower at 21 years. An issue requiring special mention is consanguineous marriages and the lack of understanding of its negative effects like high infant mortality and population decline.

Things will improve as with the 2015 push on promoting ethnic minority people in connection with post-2015 sustainable development targets, one of which is literacy rate of EM people aged 10 and over to be over 92%. However to achieve this the literacy rate must increase by 2% annually which can only be achieved by 11 EMs and the other 42 EMs will have trouble achieving hence a more targeted policy needs to be put into effect by the government. There is a call for investment in infrastructure and focus on labour market access. Women's economic empowerment and

addressing social stigmas should also be given importance.^{13 14}



Ethnic Groups in Vietnam
Source: The Nippon Foundation

Morocco

In the past five years, Moroccan authorities stopped 366,000 attempts at irregular migration to Europe; And last year it peaked at 75,000 migrants, a 6 percent increase every year.

Morocco has become a key location for thousands of refugees and migrants looking to avoid the conflicts in Africa and in the middle east like the coups d'état and the political instability in the Sahel and Sahara region and the Yemen, Syria, and Sudan conflicts. Adding on the fact that the migration routes through Libya and Tunisia are blocked, the Souss-Massa region of Morocco has become a hub for many foreign migrants.

Many migrants reported difficulties navigating and discrimination while using the public services, especially for those without the proper paperwork or passports as they are refused access to healthcare. Additionally, there is a lack of awareness among local authorities about the legal rights and services available to migrant communities. It was also likely that the typical

migration flows from the southern coasts of Morocco would be stimulated due to Morocco's attempt to support legal migration pathways via the Atlantic Initiative, which was established with the support of several Sahel countries in July 2023.

Identifying the need for integrating migration into public programs, the Collective of Sub-Saharan Communities in Morocco (CCSM) have been engaging with the authorities since 2016, with efforts doubled in the more recent years. Dedicated to fighting for minority rights, the CCSM is a migrant-led non-profit organisation that focuses on making the general public aware of the rights and culture of the migrant communities, and also advocate for inclusivity and work towards ensuring their protection and needs. The Minorities Accountability and Civic Space Programme provided support to their initiative, 'Access to Migrants in the Souss-Massa Region'.

It was clear that striking a balance between the humanitarian rights and considerations with economic and political interests is crucial, with the sudden onset of illegal migrants coming to Morocco. Addressing this at the forefront was the Global Compact for Safe, Orderly, and Regular Migration which was adopted in 2018 in Marrakech. The key focus remains in respecting human rights at the border, ensuring safe returns

¹³“Ethnic Minorities and Sustainable Development Goals: Who Will Be Left Behind?” UNDP, www.undp.org/vietnam/publications/ethnic-minorities-and-sustainable-development-goals-who-will-be-left-behind.

¹⁴Drivers-of-Socio-Economic-Development-among-Ethnic-..., documents1.worldbank.org/curated/ru/168971565786956800/pdf/Drivers-of-Socio-Economic-Development-Among-Ethnic-Minority-Groups-in-Vietnam.pdf.

and disallowing forced and collective expulsion of the migrants.^{15 16}

CONCLUSION

Cooperation among UN bodies, agencies, and mechanisms, along with their programs and activities with local governments to promote and protect minority rights, is vital to building equity, sustainable progress, and social unity. By focusing on ensuring representation, creating awareness on accessible resources based on

inclusive and specialised policies, minority groups can actively take part in their own upliftment. This helps understanding and addressing the unique challenges faced by each minority group strengthening policies and creating united societies. Thus, it is imperative to highlight and give importance to the minorities of a country for the united development of the country in all its aspects.

¹⁵Irregular Migration and Border Security in Morocco - Carnegie Endowment for International Peace | Carnegie Endowment for International Peace, carnegieendowment.org/sada/2024/06/irregular-migration-and-border-security-in-morocco?lang=en.

¹⁶“Advocating for Migrant Communities in Morocco.” *Minority Rights Group*, 18 Oct. 2024, minorityrights.org/morocco-ccsm/.

Agenda 2: Addressing Global Challenges in Water Governance

INTRODUCTION

Definition of Water Governance

Water governance refers to the overall framework and processes for decision-making and implementation of policies related to the use and management of water resources. It involves a range of actors, including central and local governments, regulators, non-governmental organisations, communities, and the private sector. Effective water governance is critical to ensure the sustainable and efficient use of water resources, address water-related challenges, and promote equitable access to water services.¹⁷

UNDP Objectives in Water Governance

As the largest UN agency in the UN family – and the largest provider of Integrated Water Resources Management (IWRM) solutions in the transboundary context – UNDP is a leading voice in water policy deliberations, while simultaneously improving lives on the ground through long-standing collaboration with governments, UN sister agencies and other partners.

UNDP addresses water challenges from the 'governance' perspective – focusing on improving the lives of the most vulnerable. UNDP supports multi-stakeholder processes and rights-based approaches at local, national and regional levels. We know that Governments must develop and implement plans that ensure equitable water access for all people while conserving this precious resource. The often-conflicting

demands and interests relating to natural resources management, require multi-level democratic and transparent processes and integrated management for their resolution. To implement agreed policy with professionalism and integrity, there needs to be accountable and competent institutions and professionals.

UNDP's Water and Ocean Governance Programme (WOGP) applies ecosystem-based approaches in supporting water governance reform, principally at the level of transboundary water management in collaboration with GEF-International Waters.

WATER AND SUSTAINABLE DEVELOPMENT GOALS

What are Sustainable Development Goals?

The Sustainable Development Goals (SDGs), also known as the Global Goals, were adopted by the United Nations in 2015 as a universal call to action to end poverty, protect the planet, and ensure that by 2030 all people enjoy peace and prosperity.¹⁸

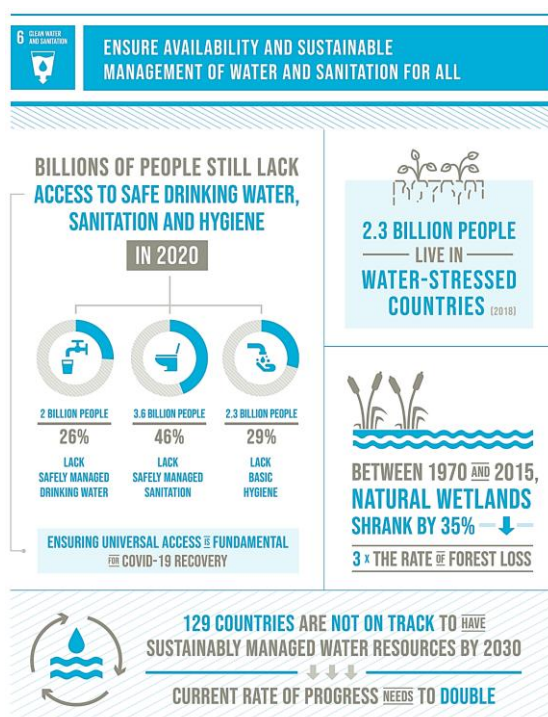
SDG 6: Clean Water and Sanitation

It is the SDG 6, that deals with the problems we are here concerned with; moreover, multiple measures by various organisations like UNDP have taken place address the same. The motto of these goals is "To ensure availability and sustainable management of water and sanitation for all". The aims of Sustainable Development Goal (SDG) 6 to be achieved by 2030 include

¹⁷Water Governance. (n.d.). United Nations Development Programme. <https://www.undp.org/nature/our-work-areas/water-governance>

¹⁸Water and Sanitation - United Nations Sustainable Development. (n.d.). the United Nations. <https://www.un.org/sustainabledevelopment/water-and-sanitation/>

ensuring universal access to safe, affordable drinking water, ending open defecation with special consideration for vulnerable groups, improving water quality by reducing pollution, increasing water-use efficiency, and protecting water-related ecosystems. The goal also seeks to expand international cooperation in water management, promote water-saving technologies, and strengthen local community involvement in water and sanitation efforts.



Progress Track of SDG 6

Source: UNADAP

Progress

None of the Sustainable Development Goal (SDG) 6 targets appear to be on track. As of 2022, 2.2 billion people were without access to safely managed drinking water. Four out of five people lacking at least basic drinking water services lived in rural areas. The situation with respect to safely managed sanitation remains dire, with 3.5 billion people lacking access to such services. Cities and municipalities have been unable to keep up with the accelerating growth of their urban populations. Deficiencies in monitoring and reporting have made it extremely challenging to

generate a thorough analysis of most other SDG 6 target indicators.

Link Between Water Governance and Other SDGs

The importance of achieving the SDG goals 6 is not merely confined to enduring access to Clean Water and Sanitation but it is expected to have a far-reaching impact on the many other goals aimed to be achieved by 2030. These include the following:

SDG 1 – No Poverty

Water governance plays a crucial role in poverty reduction by ensuring equitable access to water. When communities have reliable water sources, they can reduce health-related expenses, enhance agricultural productivity, and create job opportunities. Poor communities often bear the brunt of limited water access, incurring higher costs in time and health. Effective governance can direct investments in water infrastructure to these underserved populations, alleviating some of the burdens of poverty.

SDG 2 – Zero Hunger

Water is fundamental to agriculture, which accounts for around 70% of global freshwater use. Efficient management of water resources stabilises food production, particularly in rural areas reliant on agriculture. Governance that prioritises sustainable water practices can bolster food security in regions vulnerable to food shortages, fostering resilience in food supply chains against droughts and climate impacts. Thus, sound water governance is essential for achieving global food security.

SDG 3 – Good Health and Well-being

Effective water management is vital for public health. Access to clean water and sanitation is critical for preventing waterborne diseases,

thereby reducing the incidence of illnesses like cholera and dysentery. Water governance that emphasises sanitation and access to clean water leads to healthier populations and eases the healthcare burden on already strained systems, especially in low-income regions.

SDG 5 – Gender Equality

Water governance significantly impacts gender equality. In many regions, women and girls bear the primary responsibility for collecting water, which limits their opportunities for education and employment. Ensuring easy access to clean water alleviates this burden, allowing women and girls to participate more fully in society. Furthermore, involving women in water management decisions recognizes their unique perspectives and addresses their specific needs, promoting a more inclusive approach to resource allocation.

SDG 8 – Decent Work and Economic Growth

Sustainable water management is vital for economic growth in water-dependent sectors such as agriculture, industry, and energy. Proper governance allows these sectors to thrive responsibly, supporting livelihoods and driving economic progress. With over 70% of the global workforce engaged in water-intensive jobs, effective water management contributes to job security and stabilises economies, reducing the financial impacts of water shortages. Globally, the benefit-cost ratio of investments in water, sanitation, and hygiene (WASH) services has demonstrated considerable positive returns, particularly through co-benefits such as enhanced health, improved education, and increased employment opportunities, all while affirming basic human dignity.¹⁹

SDG 11 – Sustainable Cities and Communities

As urban centres expand, effective water management becomes increasingly critical to address risks associated with water shortages, pollution, and infrastructure challenges. Water governance frameworks that promote sustainable urban water use are essential for meeting the growing demands of cities without compromising environmental integrity. Climate resilience in water management is vital for protecting urban areas from natural hazards like floods and droughts, ultimately fostering safer and more sustainable communities.

SDG 13 – Climate Action

Water governance is integral to both climate change adaptation and mitigation. Managing water resources with climate resilience in mind helps communities withstand extreme weather events such as droughts and floods. Additionally, natural ecosystems, like wetlands, play a significant role in carbon sequestration, aiding global climate mitigation efforts. Water management policies that incorporate climate action enhance community resilience while contributing to emissions reduction.

SDG 15 – Life on Land

Effective water governance is essential for ecosystem protection. Regulated water extraction and pollution reduction support both terrestrial and aquatic biodiversity, preserving the ecosystems vital for various species. Conservation of watersheds and sustainable land use, fundamental aspects of sound water governance, ensures the health of forests, rivers, and wetlands, supporting broader biodiversity and ecological functions that sustain life on Earth.

¹⁹WASH – Water, Sanitation and Hygiene. (n.d.). UN-Water. <https://www.unwater.org/water-facts/wash-water-sanitation-and-hygiene>

SDG 17 – Partnerships for the Goals

Effective water governance relies on collaboration among diverse stakeholders. Water-related issues transcend borders and sectors, making partnerships among governments, private entities, NGOs, and communities critical for sustainable resource management. These collaborations bring together the necessary funding, technology, and expertise for impactful water governance, exemplifying global cooperation in pursuit of sustainable development goals.²⁰

CHALLENGES AND DIFFICULTIES

Water Scarcity and Distribution Inequality

The natural distribution of water resources is often characterised by high or extreme variability. Agriculture accounts for roughly 70% of freshwater withdrawals, followed by industry (just under 20%) and domestic (or municipal) uses (about 12%). However, these figures vary considerably as a function of a country's level of economic development, with higher-income countries using more water for industry whilst lower-income countries use 90% (or more) of their water for agricultural irrigation. Groundwater supplies about 25% of all water used for irrigation and half of the freshwater withdrawn for domestic purposes.²¹

Transboundary Water Conflicts

Transboundary water conflicts arise when two or more countries or regions, sharing a common water source like a river, lake, or aquifer, face

disputes over its usage, management, or conservation. These conflicts often stem from competing demands for water resources, driven by factors such as population growth, agricultural needs, industrial demands, or environmental concerns. As water flows across borders and political boundaries, managing it equitably and sustainably can become challenging, particularly when the upstream and downstream needs differ. In such cases, without cooperative frameworks, there is a risk of heightened tensions, economic strain, and potential instability among the regions involved.²²

International water law has developed principles and norms that provide the basis for transboundary water cooperation, which can also help solve disputes and contribute to regional stability. 'Water diplomacy' seeks to facilitate the political processes and practices aimed at preventing, mitigating and resolving disputes over transboundary water resources, and the development of joint water governance arrangements by applying foreign policy means at different tracks and levels. It can involve actors other than traditional state actors, including civil society organisations or academic networks. Indigenous and traditional communities may have long-standing networks across borders. Inclusive and participatory transboundary water cooperation platforms and processes lead to a common understanding of their objectives and benefits.

A large proportion of the world's freshwater resources are in transboundary aquifers. If underpinned by sound data, effective water governance and cooperation can support the

²⁰Linkage of SDGs 3, 4, 5 and 12 with SDG 6 | Department of Economic and Social Affairs. (n.d.). Sustainable Development Goals. <https://sdgs.un.org/partnerships/linkage-sdgs-3-4-5-and-12-sdg-6>

²¹Maizland, L. (n.d.). *Water Stress: A Global Problem That's Getting Worse*. Council on Foreign Relations. <https://www.cfr.org/background/water-stress-global-problem-thats-getting-worse>

²²Transboundary Waters. (n.d.). UN-Water. <https://www.unwater.org/water-facts/transboundary-waters>

conjunctive management of both transboundary surface and groundwater resources.

But the lack of transboundary cooperation has also culminated into long standing conflict across the world, including:

The Nile river Dispute: Between Egypt and Ethiopia

The construction of the Grand Ethiopian Renaissance Dam (GERD) on the Nile River has intensified tensions between Ethiopia and Egypt, with both countries framing the project in starkly different terms. For Ethiopia, the GERD represents a crucial development initiative aimed at generating hydropower and fostering economic growth. In contrast, Egypt views the dam as an existential threat, asserting that it jeopardises national water security and regional stability.²³

Since construction began in 2011, Cairo has expressed deep concerns about the potential impacts of the GERD on its water supply, which relies heavily on the Nile. Egyptian officials argue that the dam could significantly reduce water flow, exacerbating existing water scarcity issues. Meanwhile, Ethiopia emphasises that the GERD is primarily a non-consumptive hydropower project, intended to promote development and improve living conditions for its population.

Impact of Climate Change, Pollution and Degradation

Climate Change

Continued global warming is expected to exacerbate the global water cycle, leading to more frequent and severe droughts and floods.

This will result in an increase in extreme weather events, characterised by both very wet and very dry conditions. The prevalence of climate-related waterborne and vector-borne diseases is likely to rise across all regions, while freshwater ecosystems may suffer significant and often irreversible damage. The most severe impacts will disproportionately affect least developed countries, small island nations, and Arctic regions, particularly impacting indigenous communities, small-scale food producers, and low-income households.

In terms of natural hazards, floods and droughts rank among the most devastating water-related disasters. From 2002 to 2021, floods resulted in nearly 100,000 deaths (with an additional 8,000 in 2022), impacted approximately 1.6 billion people (plus another 57 million in 2022), and incurred economic losses of around \$832 billion (\$45 billion in 2022). During the same period, droughts affected over 1.4 billion people, caused more than 21,000 deaths, and led to economic losses of \$170 billion.

Water quality and pollution

As countries experience economic growth, the challenge of water pollution does not vanish; rather, it transforms. In lower-income nations, the primary issue is poor ambient water quality, largely attributed to inadequate levels of wastewater treatment. Conversely, in higher-income countries, agricultural runoff emerges as the most pressing concern. Unfortunately, comprehensive water quality data remain scarce on a global scale, particularly in the least developed regions of Asia and Africa, where monitoring and reporting capacities are severely limited.²⁴

²³Jarallah, H. (n.d.). *Nile River Conflicts – The Organization for World Peace*. The Organization for World Peace. https://theowp.org/crisis_index/nile-river-conflicts/

²⁴Maizland, L. (n.d.). *Water Stress: A Global Problem That's Getting Worse*. Council on Foreign Relations. <https://www.cfr.org/backgrounder/water-stress-global-problem-thats-getting-worse>

Insufficient Infrastructure

Water scarcity is categorised into two types: physical scarcity, caused by local ecological conditions, and economic scarcity, resulting from inadequate water infrastructure. Experts emphasise that human factors, particularly access to clean water and safe sanitation, often play a crucial role in water stress. For instance, the war in Ukraine damaged infrastructure, leaving six million people without safe water access in 2022.

The Middle East and North Africa (MENA) region faces the highest physical water stress due to low rainfall and rapidly growing urban centres. However, wealthier countries like the United Arab Emirates meet their water needs by importing food and relying on expensive desalination processes.

In contrast, countries like the Democratic Republic of Congo experience economic scarcity despite receiving ample rainfall, hindered by poor infrastructure and mismanagement. Even high-income countries like the U.S. face water stress, as seen in crises in cities like Flint, Michigan, and Newark, New Jersey, due to outdated infrastructure and rapid population growth.

Some of the major areas of water infrastructure include:

Expanding Water Supply and Distribution

Ensuring efficient and equitable water distribution is a central aim of water infrastructure policy. This involves networks of pipelines, aqueducts, and reservoirs that bring water to urban centres, industries, and remote areas. These systems, by enhancing accessibility, play a vital role in reducing disparities and

supporting economic activities across diverse regions. Moreover, with the right investments, water distribution infrastructure can improve resilience against droughts and increase public health standards.

Bolstering Irrigation Systems

Irrigation systems, particularly advanced technologies like drip and sprinkler irrigation, are essential for agricultural efficiency. Policies that support modern irrigation approaches help to minimise water wastage by delivering water directly to crops, reducing evaporation, and supporting more productive yields.

Yet, there is an important sustainability angle to consider. In areas heavily reliant on groundwater for irrigation, such as parts of South Asia and the Middle East, over-extraction is a growing problem. Unsustainable irrigation can deplete aquifers faster than they recharge, leading to long-term issues like land subsidence and soil degradation. Policymakers must consider promoting regulated groundwater use and incentivizing rain-fed agriculture or crop diversification to ease the demand on water-intensive irrigation.²⁵

Constructing Dams and Reservoirs-Sustainably

Dams and reservoirs are indispensable for multi-functional water management—whether for hydropower generation, flood control, or seasonal water storage. In addition to providing renewable energy, large dams stabilise water availability, allowing agricultural, industrial, and residential users to plan usage even during dry seasons. This reliability is especially valuable in arid regions and countries with unpredictable rainfall.

²⁵Introducing water-efficient irrigation in drought-affected areas helps communities cope with climate impacts. (n.d.). UNDP. <https://www.undp.org/kyrgyzstan/press-releases/introducing-water-efficient-irrigation-drought-affected-areas-helps-communities-cope-climate-impacts>

However, the impact of dams on river ecosystems and downstream communities is complex. Dams often disrupt natural water flows, affecting fish migration and sediment transport, which are essential for river health. Large dams can also displace communities, which requires careful and sensitive policy intervention to ensure fair compensation and resettlement options. Given these concerns, many countries are now exploring smaller-scale or “run-of-the-river” dams that offer similar benefits with fewer ecological disruptions. Including local stakeholder input and comprehensive environmental studies in planning phases can help address these issues.

Wastewater Treatment and Recycling

Wastewater treatment and recycling have become cornerstones of sustainable water management, with treated water now used in applications like irrigation and industrial processes. Advanced treatment facilities, through processes like reverse osmosis and UV disinfection, can purify wastewater for reuse, significantly reducing demand on fresh water sources. For water-scarce regions, recycling and reuse policies provide a lifeline that also supports environmental goals by reducing pollution in rivers and lakes.²⁶

Implementing Flood Control measures and Natural Infrastructure

Flood control infrastructure like levees, flood channels, and wetlands restoration is crucial for protecting communities from increasingly severe climate-induced floods. Strategic flood control reduces immediate risks to human life and property and supports regional economic stability. Moreover, “soft” infrastructure

solutions, such as wetlands, serve the dual purpose of flood mitigation and biodiversity preservation.



Integrated climate-resilient transboundary flood risk management in the Drin River basin in the Western Balkans

Source: UNDP

However, traditional hard infrastructure like levees can sometimes exacerbate flood risks downstream by blocking natural water flow, which affects both human and ecological systems. Policymakers are beginning to look toward nature-based solutions, which work with the natural landscape to absorb flood waters while supporting ecosystems. This approach aligns with sustainable development goals, but implementing these solutions requires long-term planning and careful coordination across sectors, as the impact of climate change on water flow patterns continues to evolve.

PAST INTERNATIONAL ACTIONS

United Nations Environment Programme

The United Nations Environment Programme (UNEP) plays a vital role in global water resource management. Its initiatives focus on promoting sustainable water use, protecting aquatic ecosystems, and enhancing water governance. Key among these efforts is the Global Environment Monitoring System for Water (GEMS/Water),²⁷ which provides reliable data on

²⁶Wastewater Treatment. (n.d.). UNDP. <https://sdgprivatefinance.undp.org/leveraging-capital/sdg-investor-platform/wastewater-treatment-facilities-and-technologies>

²⁷Monitoring water quality. (n.d.). UNEP. <https://www.unep.org/explore-topics/water/monitoring-water-quality>

water quality worldwide, facilitating better decision-making and tracking progress.

UNEP emphasises ecosystem-based approaches to water management through its International Waters program, which supports transboundary cooperation. This initiative fosters collaborative governance of shared water resources, aiming to reduce conflicts and promote sustainable development in regions like the Nile Basin.²⁸

The Water and Sanitation Interagency Network, co-led by UNEP, coordinates efforts among UN agencies to promote water and sanitation as part of the broader sustainable development agenda, particularly in relation to SDG 6.

Additionally, UNEP addresses the impacts of climate change on water through its Climate Change Adaptation initiatives, promoting strategies that integrate water management with climate resilience. These strategies often include enhancing natural infrastructure, such as restoring wetlands, which improve water retention and quality.²⁹

Food and Agricultural Organisation

The Food and Agriculture Organization (FAO) has played an important role in water development, focusing on sustainable water management in agriculture. One significant initiative is the Global Framework for Water Scarcity in Agriculture (WASAC), which aims to address water scarcity challenges by promoting

collaboration among various stakeholders to develop effective agricultural practices.

FAO's Aquastat program is another key resource, providing comprehensive data on water resources, usage, and management in agriculture. This information supports policymakers and researchers in understanding regional water issues and crafting targeted solutions.³⁰

In regions like sub-Saharan Africa and the Middle East, FAO has launched projects such as Smart Water Management and Water Efficient Use in Agriculture. These projects encourage the adoption of efficient irrigation methods, like drip and sprinkler systems, which help farmers maximise water use and adapt to limited availability.

International Water Treaties

The Helsinki Rules on the Uses of the Waters of International Rivers (1966)

This treaty, established by the International Law Association, set foundational guidelines for fair and equitable use of transboundary water resources. While not legally binding, they have greatly influenced subsequent water treaties.³¹

²⁸International Waters – Delivering Results. (n.d.). UNDP. <https://www.undp.org/publications/international-waters-delivering-results-o>

²⁹Climate Change Adaptation. (n.d.). UNDP. <https://www.adaptation-undp.org/>

³⁰AQUASTAT - FAO's Global Information System on Water and Agriculture. (n.d.). AQUASTAT - FAO's Global Information System on Water and Agriculture. <https://www.fao.org/aquastat/en/>

³¹International Law Association The Helsinki Rules on the Uses of the Waters of International Rivers. (n.d.). International Water Law Project. https://www.internationalwaterlaw.org/documents/intldocs/ILA/ILA-HelsinkiRules1966-as_amended.pdf

The United Nations Convention on the Law of the Non-Navigational Uses of International Watercourses (1997)

Also known as the UN Watercourses Convention, this legally binding treaty addresses the use, management, and protection of transboundary rivers, lakes, and aquifers.³²

The Berlin Rules on Water Resources (2004)

Created by the International Law Association, the Berlin Rules updated the Helsinki Rules to include environmental and sustainable development principles in water management.

The Convention on the Protection and Use of Transboundary Watercourses and International Lakes (Helsinki, 1992)

Commonly known as the UNECE Water Convention, this treaty facilitates cooperation among countries in managing shared water bodies and ecosystems in Europe, and since 2016, globally.

CASE STUDIES

India

Groundwater Management and Sustainable Agriculture in Rajasthan

Background

In India, groundwater has played a major role in sustaining the nation's economy, preserving its environment, and improving standards of living. However, the past few decades have seen an increase in groundwater extraction, with some

regions extracting more than the annual replenishment rate. Moreover, this accelerated rise in groundwater extraction has largely been unplanned and unmanaged. Although in the past some efforts were taken to promote the sustainable management of groundwater resources, in the absence of proper statutory backing to regulate groundwater use in agriculture at the national and state level, its governance remains a challenge. In this context, in 2019–20, the Government of India (GoI) launched the Atal Bhujal Yojana (ABY), a central sector scheme, to arrest the decline in groundwater levels and improve the governance of groundwater resources through effective community participation, especially in units (usually blocks) that are over-exploited (stage of groundwater development more than 100 per cent). Rajasthan faces extreme water scarcity and relies heavily on groundwater, which has been overexploited due to agriculture.³³

UNDP's Role

Through the "Groundwater Governance" project, UNDP has worked to increase community participation in managing groundwater and to promote sustainable agriculture practices. UNDP's role in Rajasthan's groundwater scheme focuses on sustainable management through community involvement, water-efficient practices, and infrastructure improvements. Key initiatives include:

Key Initiatives

Training farmers on water-efficient practices, establishing groundwater monitoring, and introducing rainwater harvesting systems.

³²Convention on the Law of Non-navigational Uses of International Watercourses (1997). (1997, May 21). United Nations - Office of Legal Affairs. https://legal.un.org/ilc/texts/instruments/english/conventions/8_3_1997.pdf

³³Gandhi, R. (n.d.). *Groundwater Management in India- Rajasthan State Report*. Rajiv Gandhi Institute for Contemporary Studies. <https://www.rgics.org/wp-content/uploads/Groundwater-Management-in-India-Study-Report-Rajasthan.pdf>

The scheme has two main components:

An institutional strengthening and capacity-building component

This is aimed at enhancing institutional capacity for effective groundwater management at all levels. This component has two parts. First, expenditure on training and capacity-building to enhance institutional capacity for effective groundwater management at all levels. Second, costs for programme management, independent verification, and, monitoring and evaluation (M&E) that include installation of equipment such as rain gauges and construction of piezometers.

Disbursement Linked Initiative

An incentive component that is aimed at rewarding or incentivising states for measures that ensure the long-term sustainability of groundwater resources.

Impact

The project has improved groundwater levels, boosted agricultural efficiency, and heightened farmers' awareness of sustainable water practices. To achieve sustainable groundwater management, the scheme incorporates several key activities. These include regularly monitoring groundwater levels, water quality, and rainfall at the panchayat level; fostering a conservation-focused mindset within the community; and coordinating the efforts of different departments involved in groundwater management.

Economically, the scheme brings substantial benefits. Agriculture and related sectors—encompassing crop production, livestock, fisheries, and forestry—form the backbone of Rajasthan's economy, employing a significant

portion of the population. According to the Directorate of Economics and Statistics (2023), these sectors contributed about 29% to the state's gross value added (GSA) at current prices in 2022–23. As these sectors largely rely on groundwater, the scheme has the potential to support Rajasthan's ongoing economic growth by ensuring a stable water supply.³⁴

Bangladesh

Community-Based Water Management for Climate Change Adaptation

Background

Bangladesh is one of the most vulnerable countries to climate change, characterised by its densely populated, coastal, and low-lying geography. The nation faces multiple serious climate risks, including rising sea levels, which threaten to inundate coastal communities and disrupt livelihoods.

River flooding is another significant risk, exacerbated by climate change and heavy rainfall during the monsoon season. Increased rainfall patterns can lead to flash floods, overwhelming local infrastructure and affecting agricultural productivity. Furthermore, the salinization of agricultural land due to rising sea levels and altered freshwater flows poses a critical threat to food security, particularly in coastal areas where many communities rely on farming and fishing for their livelihoods.

The Community-Based Adaptation (CBA) programme seeks to encourage systemic change in national adaptation-related policy through evidence-based results from a portfolio of

³⁴Khanduja, E., Chaturvedi, K., Jain, V., & Bassi, N. (2023, December 9). *Atal Bhujal Yojana Rajasthan | Participatory Groundwater Management Study*. CEEW. <https://www.ceew.in/publications/atal-bhujal-yojana-scheme-for-sustainable-participatory-groundwater-resources-management>

community-driven climate change risk management projects.³⁵

UNDP's Role

UNDP has implemented community-based adaptation projects to improve water management, especially for vulnerable coastal populations. The United Nations Development Programme (UNDP) has played a pivotal role in advancing community-based adaptation initiatives in Bangladesh, recognizing the country's acute vulnerability to climate change. Through a collaborative approach, UNDP has partnered with local governments, non-governmental organisations, and community groups to implement effective adaptation strategies that address the specific needs of at-risk populations.

Key Initiatives

Constructing resilient water infrastructure, promoting rainwater harvesting, and installing desalination plants. The CBA activities in Bangladesh consist of the following key projects:

Community-Based Wetland Project

Targeting the indigenous Chakma peoples in five villages facing declining rainfall and rising temperatures, this project aims to enhance community adaptation to climate change. It promotes sustainable crop varieties, improved agricultural practices, and water collection. Community training in alternative income-generating activities will reduce pressure on natural resources, while biodiversity conservation efforts will be supported by a newly established community committee.

Promoting Diversified Agro-Based Activities in Jamalpur District

In Bakshiganj Upazila, the Adibashi community faces challenges from increased rainfall and drought, threatening their traditional jhum cultivation practices. This project aims to address environmental degradation and livelihood risks through diversified agro-based activities that are more resilient to climate change.

Impact

These projects have improved access to clean drinking water, reduced vulnerability to climate impacts, and empowered communities to manage their water resources sustainably.

CONCLUSION

Effective water governance is crucial for addressing the complex challenges associated with water scarcity, pollution, and climate change. It requires a collaborative approach that involves all stakeholders, from local communities to national governments, ensuring equitable access and sustainable management of water resources. By integrating local knowledge and fostering innovative governance frameworks, we can enhance resilience and adaptability in the face of evolving global challenges. Ultimately, prioritising sound water governance is essential for safeguarding ecosystems, promoting sustainable development, and improving the quality of life for communities around the world.

³⁵Community-Based Adaptation: Bangladesh. (n.d.). UNDP Climate Change Adaptation. <https://www.adaptation-undp.org/projects/spa-community-based-adaptation-bangladesh>